
Determinants of Turnover Intentions: Comparing IC and IS Personnel

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Personnel is one of the most important resources for the performance of Information Systems (IS) and Information Center (IC) organizations. The scarcity of new employees, the difficulty of training and a high turnover make personnel management in these areas a difficult problem. For IS employees, the relationships between job satisfaction, organizational commitment and intention to leave the organization have been established. Because the size of the investment and the number of organizations establishing IC organizations are growing dramatically, it has become important to understand the determinants of turnover intentions for IC as well as IS employees. Are IC employees similar to their IS counterparts? Or, is their nature basically different, as some studies have suggested? This study examines the differences between IS and IC employees in terms of demographic characteristics, participation on boundary spanning activities, role stressors, overall job satisfaction, organizational commitment and turnover intentions. The differences are found to be significant and call for special attention from IC managers to manage more properly their personnel resources. IC employees were found to participate more extensively in boundary spanning activities, experienced more role stressors (role ambiguity and role conflict), were less satisfied with their jobs and less committed to their organization. The findings also demonstrate the importance of organizational commitment as an intervening variable in models of turnover. While overall job satisfaction had both direct and indirect effects on turnover intentions among IC employees, for IS personnel it had only indirect effects through organizational commitment. The effects of role stressors and boundary spanning activities were found to be indirect via overall job satisfaction and organizational commitment for both IC and IS employees. The implications of these findings for practicing managers and for future research are discussed.

Information Center—IC Personnel—IS Personnel—Turnover intentions—Job satisfaction

Introduction

The importance of qualified and motivated personnel to the success of Information Systems (IS) departments is widely recognized (Baroudi 1985). Hiring and training have been considered critical challenges for IS managers in the past (Ives and Olson 1981). In general, IS personnel management is likely to become even more difficult due to the perennial turnover problem (Goldstein and Rockart 1984, Laudon and Laudon 1988), the difficulty of personnel training (Bartol and Martin 1982), and the chronic shortage of new people entering the field (Couger 1990). IS professionals have historically displayed a high rate of turnover that reached nearly 20 percent in 1987 (Connolly 1988), and is about twice the average of business managers and professionals (Ludlum 1988). Excessive turnover in the IS area can be dysfunctional to organizations because of the shortage of experienced employees and the high cost of training new hires (Goldstein 1989, Lucas 1989).

The advent of end-user computing is likely to exacerbate personnel management problems for most organizations. The proliferation of microcomputers among users who are not information systems professionals and the growth of mainframe based end-user computing have been widely reported (Carr 1987; Guimaraes and Ramanujam 1986; Guimaraes 1984; Igbaria, Pavri and Huff 1989). When *Computerworld* surveyed the 100 organizations rated as having the most effective use of computerized information systems, these organizations were already found to have on the average 35 PC/workstations per 100 employees, with the top 25 organizations having an average of 44 PC/workstations per 100 employees (Sullivan-Trainor 1988). As end-user computing becomes pervasive in most organizations, its diversity grows in terms of types of applications, types of end-users, levels of end-user computer literacy, etc. Contrary to early expectations, end-users do not become independent; instead, they increasingly demand better equipment, more training, coaching, consulting, technical support, etc. To provide this collection of services, organizations will have to hire or groom a special kind of support personnel, easily able to cross departmental barriers, willing and able to communicate with a wide variety of end-users at different computer literacy levels trying to address a plethora of diverse business problems.

The expansion in end-user computing requires substantial investment in personnel and facilities for support (Guimaraes 1986; Leitheiser and Wetherbe 1986). The large number of organizations that have established Information Centers (IC) attests to their importance to support end-user computing activities. The American Management Association (1988) reported that 58 percent of the large companies surveyed already had a discrete unit to support end-user computing.

Very clearly, ICs have evolved over time (Guimaraes 1984) and are continuing to change. A survey by Crwth Computer Courseware has 65 percent of respondents reporting their IC's role changing in some way. Respondents are evenly divided about whether their IC's role is rising or decreasing. Forty percent reported some functions being shifted to other parts of the organization and that training, product specialists and application development are understaffed and overworked. Most organizations (58 percent) are keeping active IC organizations, 6 percent have disbanded their IC as inappropriate to their needs, 34 percent do not have one and presently have no plans for one, and 2 percent will establish an IC in the near future (Crwth 1991, p. 10).

Some of the burden for end-user computing support is increasingly falling on IS and user departments, away from understaffed centralized IC (Crwth 1991, p. 13). Similar interpretation of the Crwth survey is reported by Roberts (1991), and a dramatic impact of end-user computing on IS department objectives and organization is also being reported by others (Hildebrand 1991, Juneau 1991). However, regardless of how a company chooses to organize to provide end-user computing support, the services must be available and require considerable crossing of departmental boundaries and special skills. As discussed later in the sample and procedure section, in this study the term "IC" is broadly defined to include all personnel whose primary function is to support end-user computing (i.e., coaching, help desk). In this context, end-users have a greater and more direct control over systems development. These new personnel roles may or may not be performed through a formal organizational subunit within IS departments. On the other hand, IS personnel perform traditional systems development activities (i.e., analysis, programming) with end-users being secondary participants in the process.

The importance of gaining a better understanding of the factors related to the recruitment, development, and retention of IC professionals is underscored by the high demand and the extremely, and apparently chronic, short supply of IC personnel (Carr 1987, 1988; Connolly 1988; Rainer, Carr, Snyder and Frolick 1989; Saks 1989). More recently, the shortage of IC personnel can be seen in the shift, within many organizations, of end-user computing support tasks to IS and end-user departments because the central "IC's are understaffed and by inference, overworked" (Crwth 1991, p. 11). Further aggravating the problem are the rising personnel costs relative to equipment costs (Ahituv and Neumann 1990), and the high rates of turnover among IS employees in general, and IC staff in particular (Bartol and Martin 1982, Lucas 1989, Rifkin 1985).

The importance of effective human resource management is also affected by the fact that personnel expenses consumed an average of 43 percent of the IS budget in 1988, with no expected changes in the foreseeable future (Lockwood and Sobol 1989). One important aspect of effective human resource management within IS is the control of employee turnover (Baroudi 1985, Bartol 1983, Bartol and Martin 1982, Laudon and Laudon 1988). In order to reduce excessive turnover among IS employees, it is necessary first to understand the reasons behind the turnover. To the extent that specific factors contributing to IS turnover can be controlled by the organization, IS employees' retention could be increased through appropriate administrative actions designed to address the problem areas.

Given the sensitivity of the current turnover problem facing the IS profession, it is surprising that so little research has been conducted to identify the reasons for high turnover among IS employees and define effective policies to manage it. Perhaps the most comprehensive studies in this area were conducted by Baroudi (1985) and Bartol (1983). Although these studies have provided a more complete understanding of the variables related to turnover among IS employees, additional research is needed to synthesize the findings and to formulate a parsimonious model that explains the process by which IS employees intend to leave the organization. There is also a need to re-examine the antecedents of turnover intentions among IC employees and whether these kinds of antecedents for IS employees reported by Baroudi (1985) are applicable to IC employees.

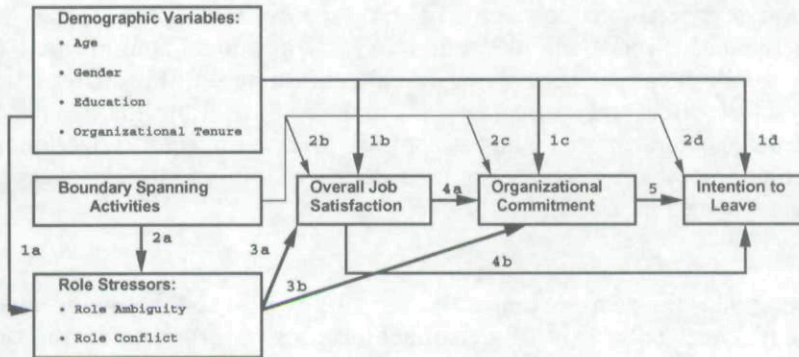
There is considerable evidence that IC and IS employees differ on a number of demographic, personality and motivational factors. It has been noted that IC personnel, compared to their IS counterparts, have greater need for interpersonal skills and social interaction; have higher business skills and lower technical skills; tend to be more creative and persistent; and tend to be more end-user/people oriented (Carr 1987, Gunton 1987, Lucas 1989, Perry 1987, Rainer et al. 1989). In addition, it has been reported that over half of IC staff members are women, compared to one-third of IS personnel (Johnson 1990). Nearly half of the IC professionals are 30 years of age or younger; and a large majority are better educated than their IS counterparts (Carr 1988, Rainer et al. 1989). There is also evidence that IC personnel face limited advancement, particularly into senior positions within IC organizations (Gunton 1987, Perry 1987). The latter, coupled with negative salary differentials, are thought to be responsible for some IC personnel job dissatisfaction and eventually high turnover (Carr 1988; Rainer et al. 1989). The widespread perception of a lack of career path, and a lack of formal promotion procedures, pose a threat to job security and are likely to motivate IC personnel to leave the organization (Carr 1988; Gunton 1987; Perry 1987; Rainer et al. 1989). These findings have major implications for the management of IC (Carr 1988) and emphasize the need to examine the determinants of turnover for IC personnel.

The primary purpose for this study is to examine the differences between IC and IS employees in terms of demographic variables, boundary spanning activities, role ambiguity, role conflict, job satisfaction, organizational commitment, and intention to leave the organization. Further, the study uses a comprehensive multivariate analysis model to examine the complex network of relationships among these variables and to identify any differences between IC and IS employees. Specifically, the empirical test of the proposed model is designed to determine: (1) whether there are differences in the demographic characteristics, boundary spanning activities, role stressors, job satisfaction, organizational commitment and intention to leave among IC and IS employees; (2) the pattern of relationships among the antecedents of turnover intentions; and (3) whether job satisfaction and organizational commitment mediate the effects of the exogenous variables on turnover intentions of IC and IS employees. The next section presents the theoretical basis for this study. The first part of the section will present a discussion on the differences between the IC and IS employees, and the second part will present the path model as tested separately for IC and IS employees.

The Conceptual Framework

As mentioned earlier, turnover intentions of IS personnel have been studied by Bartol (1983) and Baroudi (1985). The well-established relationships between overall job satisfaction, organizational commitment, and turnover intentions clearly provide the basis for understanding the antecedents of turnover intentions (Baroudi 1985, Goldstein 1989, Goldstein and Rockart 1984). For this study, the model proposed and tested by Baroudi (1985) to study the turnover intentions of IS professionals was modified slightly to incorporate the demographic variables. This modification is designed to support a more fine-grained analysis of the relationships between demographic variables, boundary spanning activities, role stress, job satisfaction, organizational commitment, and turnover intentions among IS and IC personnel.

Figure 1 presents the conceptual model used in this study. The model was separately tested for IC and IS employees. This model is built on the theoretical perspec-



Note. The numbered arrows correspond to the hypotheses described in the body of the paper.

FIGURE 1. The Determinants of Turnover Intentions.

tives provided by turnover theorists (Mobley, Griffeth, Hand and Meglino 1979), and on recently proposed frameworks and empirical findings regarding careers among IC and IS employees (Baroudi 1985; Bartol 1983; Carr 1987, 1988; Gunton 1987; Perry 1987; Rainer et al. 1989). The model includes four sets of variables: (a) demographic variables, (b) boundary spanning activities, (c) two role stressors: role ambiguity and role conflict, and (d) career outcomes: overall job satisfaction, organizational commitment and intention to leave.

Demographic variables include age, gender, education and organizational tenure. Boundary spanning activities relate to individuals crossing departmental or organizational boundaries in order to perform their job duties (Adams 1976; Baroudi 1985). In the information systems field, boundary spanning activities are common among systems analysts and end-user support staff (IC personnel). They are frequently required to interact with end-users to define and assist with information requirements and end-user computing problems.

Role ambiguity has been defined as the degree to which the employee lacks clear information regarding (a) the expectations associated with a role, (b) methods of fulfilling known role expectations, and/or (c) the consequences of role performance (Kahn, Wolfe, Quinn, Snoek and Rosenthal 1964). Role conflict is defined as incompatibility and incongruity in the expectations associated with a particular job role. Job satisfaction refers to the primary affective reactions of individuals to various facets of the job and job experiences (Locke 1976). Organizational commitment reflects one's evaluation of the organization as a whole, and encompasses three dimensions: (a) a strong belief in and acceptance of the organization's goals and values, (b) willingness to exert considerable effort on behalf of the organization, and (c) a strong desire to maintain membership in the organization (Porter, Crampon and Smith 1976). Intention to leave refers to the individual's perceived likelihood that s(he) will be staying or leaving the employer organization. The differences between the IC and IS employees will be examined in terms of the above variables.

The model in Figure 1 posits that (1) demographic variables will be directly related to role stressors, job satisfaction, organizational commitment, and turnover intentions; (2) boundary spanning activities will be directly related to role stressors, job

satisfaction, organizational commitment, and turnover intentions; (3) role stressors will directly affect overall job satisfaction and organizational commitment; (4) job satisfaction will directly affect organizational commitment; (5) job satisfaction will inversely affect (direct and indirect effects) employees' intentions to leave the organization; and (6) organizational commitment will inversely affect turnover intentions. In addition, these effects will be examined separately for IC and IS employees and the differences between IC and IS will be examined.

The Expected Differences Between IS and IC Personnel

As said earlier, the primary objective of this study is to test the major hypothesis that IC employees differ from IS personnel in terms of turnover intention and its antecedents. Specifically, the following hypotheses comparing IC personnel to their IS counterparts are to be tested.

The IS literature has reported that IC personnel are likely to be women, much younger, more educated and have a shorter tenure with the organization than the general IS counterparts (Carr 1988; Rainer et al. 1989). In addition, it has been shown that IC people have continuous interaction with individuals both internal and external to the IC, requiring considerable intraorganizational boundary crossings (Christoff 1990; Perry 1987). Therefore, we expect that the level of boundary spanning activities among the IC staff will be much higher than among their IS counterparts.

Role clarity was proposed as one of the most critical factors affecting the success of IC (Magal et al. 1988). Role ambiguity for IC people is expected to be higher than for IS people for several reasons. Compared to IS professionals (development), IC people (user support) are likely to face a wider variety of end-users and end-user problems as end-user computing proliferates throughout the organization. End-users are likely to differ more widely in various ways. Before end-user computing, computer illiteracy was considered a common occurrence. Today, many end-users know more about some aspects of computing than IS professionals or IC specialists. For example, within a given company many times end-users initiate end-user computing activities (i.e., systems development, operations) before an IC is even established. In many cases, the IC staff has to catch up with some end-users while having to hold hands and motivate the slower ones. Quite often, IC personnel must draw a fine line between all-out support of end-user requests and partial support or merely providing direction for end-users. In some cases, the end-user expectations may be intentionally vague—do what you like as long as you solve my problem. Furthermore, the unstructured nature of work and task, the demand for new and more sophisticated services, and the lack of resources to meet that demand, may lead to vague expectations about what should be done and create a confusing environment for the IC staff (Carr 1988). In the case of IS, the roles and types of service provided have been more clearly defined over the years. Therefore, it is hypothesized that IC personnel will, on the average, report higher role ambiguity than IS employees.

IC employees are likely to have higher role conflict because frequently they work under dual supervision. On one hand, they formally report directly to the IC manager. On the other hand, individual end-users and departments have considerable influence over IC personnel (Christoff 1990; Perry 1987). In many cases, end-users formally evaluate the support they get and often try to countermand the directives of the IC manager and to change the priorities by which IC employees respond to end-user requests. IC employees many times must handle end-users requests which

require special and quick response. While an individual end-user may think that his or her request is the most important, other end-users tend to disagree. That may cause more role conflict for the IC employee. The IC employees may also find themselves caught between the end-user computing spirit of *laissez faire* and the more systematic/disciplined view of computing sponsored by IS departments which are responsible for managing and controlling computing resources in the organization. Therefore, it is hypothesized that IC personnel will, on the average, report higher role conflict than IS employees.

Given the variable linkages proposed in the model (Figure 1), and the expectation that IC personnel will report higher role ambiguity and role conflict, we predict that IC employees will show lower job satisfaction. In turn, we expect them to show lower commitment to their organizations and stronger intention to leave. Therefore, it is hypothesized that IC personnel will, on the average, report lower job satisfaction and organizational commitment, and higher intention to leave the organization compared to IS employees.

The Determinants of Turnover Intentions

Demographic Variables. A number of empirical studies indicate that demographic variables are relevant to understanding the turnover process because they are related to important job satisfaction, organizational commitment and employee's turnover intentions (Arnold and Feldman 1982; Bluedorn 1982; Williams and Hazer 1986). Age and organizational tenure have been found to be positively related to job satisfaction and organizational commitment (Arnold and Feldman 1982; Cotton and Tuttle 1986; Price 1977). The relationship of gender to job satisfaction and organizational commitment remains unclear, with some studies reporting that males are more satisfied and committed than females (Arnold and Feldman 1982), and others reporting no gender differences (Brooke and Price 1989). Education has also been found to be negatively related to satisfaction (Parasuraman 1989) and organizational commitment (Bluedorn 1982). Relationships of gender and education to turnover have also been observed, with women and highly educated employees more likely to leave the organization than men and less educated employees (Arnold and Feldman 1982; Cotton and Tuttle 1986). Moreover, prior research suggests that demographic variables may have direct effects on turnover intentions over and above their effects on intentions through satisfaction and commitment (Parasuraman 1989).

The relationships between the demographic variables and role stressors were also examined. Education was found to be very weakly and positively related to role ambiguity and role conflict (Brief, Aldag, Van Sell and Melone 1979). Tenure, on the other hand, tends to be slightly negatively correlated with role ambiguity and unrelated to role conflict (Curtiss, Hammel, Heinen and Johnson 1978). It is suggested that the longer one is on the job, the more information he or she obtains and the less role ambiguity in the job. Age was also included in this study; however, Jackson and Schuler (1985) stated that the correlations of age and role stressors are always very weak, occasionally significant, and almost always negative. Finally, Tung (1980) reported that women experienced lower levels of role stress than men. However, others have found higher prevalence of role ambiguity among women than men (Zalesnik, Kets de Vries and Howard 1977). Thus, age, organizational tenure, gender, and education are examined as salient demographic variables potentially predictive of role stressors, overall job satisfaction, organizational commitment and turnover

intentions. Hypothesis Set 1 predicts that those demographic variables have direct effects on role stressors, job satisfaction, organizational commitment, and turnover intentions. Thus we propose:

H1a: *Demographic characteristics are directly related to role stressors.*

H1b: *Demographic characteristics are directly related to overall job satisfaction.*

H1c: *Demographic characteristics are directly related to organizational commitment.*

H1d: *Demographic characteristics are directly related to turnover intentions.*

Boundary Spanning Activities. Boundary spanning activities require individuals to cross departmental or organizational boundaries in performing their job duties (Adams 1976; Baroudi 1985; Lysonski 1985). Recent literature (Ahituv and Neumann 1990; Lucas 1989) indicates that the process of system analysis and design requires continual interaction with individuals both internal and external to the IS department, necessitating the crossing of many organizational boundaries. Specifically, IC employees are required to provide help and assistance in designing and debugging a wide variety of user applications and tools (Carr 1988). To accomplish that, they must cross departmental boundaries and interact extensively with end-users, IC applications developers, and vendors, software and hardware and services.

It is not clear whether participation in boundary spanning activities will have a positive or a negative impact on turnover intentions. On one hand, boundary spanning can produce extensive role conflict (Baroudi 1985; Lysonski 1985) and may generate role ambiguity (Lysonski and Woodside 1989), suggesting a positive relationship between boundary spanning and turnover intentions. On the other hand, participation in boundary spanning activities has been found to increase the level of employee satisfaction and strengthen the individual's organizational commitment (Baroudi 1985; Lysonski 1985; Lysonski and Woodside 1989), that would suggest a negative relationship between boundary spanning and turnover intentions. Although the sign of the relationship cannot be specified on theoretical grounds, Hypothesis Set 2 predicts that boundary spanning activities have direct effects on role conflict, role ambiguity, job satisfaction, organizational commitment, and turnover intentions. In this case the hypotheses are:

H2a: *Boundary spanning activities are directly related to role stressors.*

H2b: *Boundary spanning activities are directly related to overall job satisfaction.*

H2c: *Boundary spanning activities are directly related to organizational commitment.*

H2d: *Boundary spanning activities are directly related to turnover intentions.*

Role Stressors. The two role stressors (role ambiguity and role conflict) were reported to be negatively associated with job satisfaction and organizational commitment, and positively related to turnover intentions among IS employees and others (Baroudi 1985; Bedeian and Armenakis 1981; Brooke, Russell and Price 1988; Goldstein and Rockart 1984; Miles and Perreault 1976; Van Sell, Brief and Schuler 1981). These studies suggested that the effect of role ambiguity and role conflict on turnover is mediated through intervening job satisfaction and organizational commitment. As

a result of insufficient information to perform the job adequately, unclear expectations of peers, ambiguity of performance evaluation methods, extensive job pressures, and lack of consensus on job functions or duties among end-users, technical people and supervisors, employees may feel less satisfied with the job, less committed to the organization, and display a greater propensity to leave the organization. Therefore, Hypothesis Set 3 predicts that role conflict and role ambiguity have indirect effects on turnover intentions through job satisfaction and organizational commitment. It is expected that role conflict and role ambiguity are negatively related to satisfaction and commitment. In this case the hypotheses are:

H3a: *Role stressors are directly related to overall job satisfaction.*

H3b: *Role stressors are directly related to organizational commitment.*

Overall Job Satisfaction. Several studies have focused on the relationships between job satisfaction, organizational commitment and turnover intentions. Moderate zero-order correlations in the ranges of 0.30 to 0.62 between job satisfaction (particularly overall job satisfaction or satisfaction with the work) and organizational commitment have been observed repeatedly in the IS field as well as in other fields (Baroudi 1985; Bartol 1983; Brooke et al. 1988; Parasuraman 1989). It is reasonable to expect that high levels of job satisfaction would enhance organizational commitment, since employees who are satisfied with their jobs should perceive greater benefits in retaining membership in their organizations than employees whose jobs have been less gratifying. Job satisfaction and organizational commitment were reported to be related but represent two distinct constructs in the IS literature (Baroudi 1985; Bartol 1983; Goldstein and Rockart 1984). Mowday et al. (1982) argued that organizational commitment focuses on the organization as a whole rather than on the specific job. Job satisfaction refers to the primary affective reactions of individuals to various facets of the job and job experiences. The findings of Dougherty, Bluedorn and Keon (1985), Cotton and Tuttle (1986), and Michaels and Spector (1982) provide evidence that job satisfaction has a direct effect on turnover intentions, as well as an indirect effect through organizational commitment. That contradicts the intuitive notion that job satisfaction affects intention to leave primarily through its effect on organizational commitment, as reported by Baroudi (1985). Therefore, Hypothesis Set 4 predicts that job satisfaction has direct effects on organizational commitment and turnover intentions, and an indirect effect on employee turnover intentions via organizational commitment. Thus, in this case the hypotheses are:

H4a: *Overall job satisfaction is directly related to organizational commitment.*

H4b: *Overall job satisfaction is directly related to turnover intentions.*

Organizational Commitment. The importance of organizational commitment as a key attitudinal variable contributing to intention to leave is well documented in the literature (Baroudi 1985; Bluedorn 1982; Parasuraman 1989; Porter et al. 1976). Employees who are highly committed to their organization are less likely to leave than employees who are relatively uncommitted. It was also reported that organizational commitment is more closely related to turnover intentions than job satisfaction (Baroudi 1985; Bluedorn 1982; Shore and Martin 1989). Therefore, Hypothesis

5 predicts that organizational commitment has a direct effect on turnover intentions. Hence, we have:

H5: *Organizational commitment is directly related to turnover intentions.*

Method

Sample and Procedure

Thirty-eight companies with operations in the state of Ohio, known to have significant IC activities, were asked to participate in this study. Each of the companies had a minimum of five years of end-user computing experience. The IS director, the source of contact, introduced the researchers to IC managers or other IC personnel, and IS employees (systems development) invited to participate in the study. Twenty-eight companies agreed to participate. Among these, 14 companies were in manufacturing, 6 in multi-industry, 2 in utilities, 2 in insurance, and the remaining included banks and others.

Ninety-two IC employees and 140 IS employees (5 per company) from the 28 companies were asked to participate in the study. In this case, IC employees were defined as those whose primary duties are to support and manage end-user computing activities in the organization. Their primary duties were coaching, training, equipment demonstration and technical support. The IS employees mostly perform system development (programmer/analyst) duties. Participation was voluntary, and participants were assured that their individual responses would be treated as confidential. A questionnaire was distributed to each participating IC and IS employee. Seventy-six IC employees and 133 IS employees completed the questionnaire (82.6 and 95 percent response rates for IC and IS, respectively). The final sample consisted of 209 individuals.

Of the 209 respondents, 99 (47.4 percent) were women and 110 (52.6 percent) were men. The age of the participants ranged from 21 to 54, with an average of 31.97 years. Their average length of service or organizational tenure was 4.31 years. Because organizational tenure and age are not normally distributed, a logarithmic transformation has been used (Velleman and Hoaglin 1981). The educational level of the participants included completed high school (20.6 percent), college degree (64.1 percent), and master degree (15.3 percent).

Measures

Demographic Variables. The analysis included five demographic variables: employee type, gender, age, organizational tenure, and education. Employee type was coded (1 = IC; 2 = IS) on the questionnaire cover sheet. Gender was assessed with a fixed-response item (1 = female; 2 = male). Age was measured in years. Education consisted of four levels from (1) high school to (4) Ph.D. degree. Organizational tenure included length of time employees worked for the company. In addition to demographic variables, several multiple-item measures were used to operationalize the antecedents of turnover intentions.

Boundary Spanning Activities. Boundary spanning activities, defined as communication beyond an employee's immediate department, were measured by modifying the ten-item instrument developed by Miles and Perreault (1976) and used by Baroudi (1985). The measure is useful to investigate several aspects of boundary spanning activities, including internal verbal and written communication, formal meetings and written communication, and informal contact and written communication.

Each item was scored on a five-point scale ranging from (1) not part of my job to (5) a very significant part of my job. The internal consistency reliability of the scale as used in the current study was 0.83.

Role Stressors. Role ambiguity and role conflict were operationalized by the scales developed by Rizzo, House and Lirtzman (1970). Each scale was scored using a seven-point response mode ranging from (1) "very false" to (7) "very true." The responses were averaged across relevant items. The greater the score, the greater the perceived stress. The internal consistency reliability (alpha coefficient) of the measures of role ambiguity and role conflict were 0.84 and 0.78, respectively.

Overall Job Satisfaction. Satisfaction was assessed through the Job Descriptive Index (JDI) developed by Smith, Kendall and Hulin (1969). The JDI (Smith et al. 1969) was selected because it is viewed by many investigators as one of the most thoroughly researched and developed measures of its kind (Campbell 1970; Vroom 1964). In terms of both predictive power and construct validity, the five components of job satisfaction have shown strong relationships with other important variables such as organizational commitment and the employee's intentions to leave. The JDI measures satisfaction with five facets of the job: the work itself, supervision, co-workers, pay, and promotion. The three-point JDI responses were defined so that a negative response ("yes" to a negative item or "no" to a positive item) was scored 0. A positive response was scored 3, and "I don't know" response ("??") was scored 2. The coefficients of reliability (Cronbach's Alpha) for each of the five subscales are: work (alpha = 0.84), supervision (alpha = 0.91), coworkers (alpha = 0.88), pay (alpha = 0.85), and promotion (alpha = 0.91). The items within each of these five components were then summed and averaged to form submeasures of job satisfaction. Since most of the IS research focused on overall job satisfaction, a factor analysis (Nunnally 1978) was conducted to test for underlying homogeneity in the five subscales of job satisfaction. The results showed a single factor solution (eigenvalue of 2.92), which explained 58.4 percent of the explained variance. Therefore, the five subscales were aggregated and averaged to create a composite measure of overall job satisfaction. The items were recorded such that high scores reflected high level of overall job satisfaction. The alpha reliability coefficient for the overall job satisfaction scale was 0.82.

Organizational Commitment. It is defined as the employee identification with a particular organization and the desire to maintain membership. It is measured by an abbreviated version of the Organizational Commitment Question (OCQ) developed by Porter et al. (1976). The nine items used to construct the scale tap two of the three dimensions of commitment included in the longer version of the OCQ: (1) a strong belief in and acceptance of the organization's goals and values; and (2) a willingness to exert considerable effort on behalf of the organization. In order to avoid concept redundancy (Morrow 1983), the six items reflecting a strong desire to maintain membership in the organization were excluded from the measure of commitment because they overlapped with the measure of turnover intentions. Thus the shorter version of the scale used in this study represents a more "pure" measure of the affective dimensions of commitment to the employing organization. The response options to the items ranged from (1) strongly disagree to (5) strongly agree. The reliability and validity of the nine-item version have been found to be acceptable (Brooke, Russell

and Price 1988). The items were recorded such that high scores reflected more commitment to the organization. The reliability coefficient (α) for this scale was 0.87.

Intention to Leave. Intention was measured via three items developed by Mobley, Horner and Hollingsworth (1978), and assessed the employee's intent to leave the organization. Responses to each item were recorded on a five-point scale ranging from (1) "very unlikely" to (5) "very likely." The three items were summed and averaged to form the intention to leave index (Cronbach's α coefficient = 0.89).

Data Analysis Procedure

Analysis of variance (ANOVA) procedure was conducted to test whether there are differences among the study variables. Chi-square tests of independence were used to determine whether the nominal demographic variables (gender and education) were systematically associated with type of employee (IC or IS). Table 2 shows that there was a significant difference in organizational tenure, where the IS employees were found to have a longer tenure with the organization than the IC counterparts. Therefore, it became necessary to control for organizational tenure in all analyses to ensure that conclusions regarding employees' differences were not confounded by differences in their organizational tenure. Differences between IC and IS employees in the boundary spanning activities and role stressors were examined with the analysis of covariance (ANCOVA). The independent variable was employee type, the covariance was organizational tenure, and the dependent variables were boundary spanning activities and the two role stressors. We also examined the relationships between job satisfaction, organizational commitment, turnover intentions, and type of employee. In this case, we controlled for the predictors of all the dependent variables (for example, we controlled for demographic variables, boundary spanning activities, and role stressors in the analysis of overall job satisfaction) using ANCOVA, where the independent variable was employee type, the covariates were the predictors, and the dependent variables were job satisfaction, organizational commitment and intention to leave.

Path analysis was used to determine whether the observed pattern of relationships among the variables was consistent with the model presented in Figure 1. The path model was tested separately for the IC and IS employees. The choice of path analysis over structural equation modeling techniques such as LISREL is based on the small sample size. However, several steps were taken to check for possible violations of the assumptions underlying path analysis (Billings and Wroten 1978; Heise 1969). Examination of the alpha reliability coefficients indicated satisfactory levels of internal consistency reliability among the multiple-item scales (i.e., $\alpha \geq 0.70$). Moreover, the intercorrelations among the study variables revealed no evidence of extreme multicollinearity (i.e., $r \geq 0.80$). Finally, the residuals of the endogenous variables were tested for autocorrelation (Makridakis and Wheelwright 1989) using the Durbin-Watson "d" test. The values of the "d" statistics were between 1.79 and 2.09, implying an absence of autocorrelation among the residuals. Note that, for a medium effect size, the statistical power for the IS group exceeded 0.80. However, the statistical power for the IC group was low (but exceeded 0.50). For a large effect size, the statistical power for both subsamples exceeded 0.80 (see Cohen 1988 for more detail).

The omitted parameter test (James, Mulaik and Brett 1982) was used to determine whether the paths proposed in Figure 1 were statistically significant, and whether or not the unpredicted paths were significant. This test involves checking all the direct

paths (predicted or unpredicted) among the model variables. In this manner, one identifies all possible ways to corroborate or contradict the proposed model.

Hierarchical multiple regression analyses (Cohen and Cohen 1983) were performed for the omitted parameter test, and to assess the direct, indirect and common cause effects (Kerlinger and Pedhazur 1973) of the independent variables on job satisfaction, commitment, and turnover intentions (dependent variables). First, each role stressor was separately regressed on demographic variables and boundary spanning. Next, overall job satisfaction was regressed on demographic variables, boundary spanning activities, and role stressors. Organizational commitment was regressed on demographic variables, boundary spanning, role stressor and overall job satisfaction. Finally, intention to leave was regressed on demographic variables, boundary spanning, role stressors, overall job satisfaction, and organizational commitment. The analysis of variance was used to examine the relationships between type of employee and demographic variables, boundary spanning activities, and the two role stressors. In addition, all the above analyses were separately conducted for IC and IS employees.

A direct effect is the impact of a variable presumed to be a cause of another variable, measured by standardized regression coefficients (beta). The final beta coefficient (after all independent variables have entered the analysis) represents the direct effect of the variable. An indirect effect represents the effects of antecedent variables mediated by the intervening variables in the model, i.e., role stressors, overall job satisfaction and organizational commitment. It is the product of the path coefficients (direct effects) along an indirect route from cause to effect via tracing arrows in the headed direction only. When more than one indirect effect exists, their sum is the total indirect effect (Ross 1975). The common cause effect indexes the amount of spurious covariation between two variables caused by other prior variables in the path model. It is the product of the path coefficients (direct effects) along both "legs" of the common cause effect. If there is more than one common cause variable, the total common cause effect is their sum. The original correlation between the predictor variables and the dependent measures were compared with the recomputed correlations represented by the sum of the direct, indirect and common cause effects. The effects not explained by the path model is the difference between the simple correlation " r " and the sum of the direct, indirect and common cause effects (Billings and Wroten 1978; Kerlinger and Pedhazur 1973). If the path model efficiently captures the relationships specified among the variables, the unexplained effects should be near zero.

A more conservative way to assess differences between IC and IS employees on the determinants of turnover intentions is to examine the significance of the differences between the beta weights obtained for IC and IS, respectively. That was done by applying the formula proposed by Cohen and Cohen (1983).

Results

Results for Hypotheses Testing

Table 1 presents the means, standard deviations, alpha coefficients, and matrix of intercorrelations among the variables examined in this study. The results of the analysis of variance and Chi-square test are presented in Table 2. The univariate analyses showed that IC employees, when compared to IS employees, reported higher level of boundary spanning activities (IC = 2.69 vs. IS = 1.90, $F = 86.87$, $p \leq 0.001$),

TABLE 1
Matrix of Intercorrelations among Study Variables (N = 209)

Variables	Mean	S.D.	Alpha	1	2	3	4	5	6	7	8	9	10	11
1. Type of Employee (1 = IC; 2 = IS)	1.64	0.48	NA	1.00										
2. Age*	1.51	0.79	NA	-0.04	1.00									
3. Gender (1 = Female; 2 = Male)	1.53	0.5	NA	0.08	-0.07	1.00								
4. Education	1.95	0.60	NA	-0.04	0.13	-0.09	1.00							
5. Organizational Tenure*	0.46	0.37	NA	0.29	0.32	-0.08	-0.13	1.00						
6. Boundary Spanning Activities	2.19	0.70	0.83	-0.54	0.13	-0.02	-0.03	0.26	1.00					
7. Role Ambiguity	3.13	0.95	0.84	-0.14	-0.10	0.06	0.08	-0.06	0.01	1.00				
8. Role Conflict	4.03	0.96	0.78	-0.26	-0.13	0.08	0.08	0.08	0.15	0.33	1.00			
9. Overall Job Satisfaction	1.77	0.45	0.82	0.31	0.01	-0.04	-0.06	0.22	0.10	-0.46	-0.36	1.00		
10. Organizational Commitment	4.46	1.27	0.87	0.26	0.01	-0.02	-0.08	-0.18	0.15	-0.17	-0.09	0.44	1.00	
11. Intention to Leave	2.71	1.04	0.89	0.06	-0.13	0.04	0.07	0.03	-0.17	0.33	0.17	-0.23	-0.20	1.00

* Using the log of age and organizational tenure.

Notes: The absolute value of correlations ≥ 0.11 are significant at 0.05 or better, and correlations ≥ 0.15 are significant at 0.01 or better.

higher level of role stressors [for role ambiguity (IC = 3.31 vs. IS = 3.03, $F = 4.24$, $p \leq 0.05$) and role conflict (IC = 4.35 vs. IS = 3.85, $F = 14.41$, $p \leq 0.001$)], lower levels of overall job satisfaction (IC = 2.12 vs. IS = 2.34, $F = 22.29$, $p \leq 0.001$), and lower organizational loyalty and commitment (IC = 3.94 vs. IS = 4.57, $F = 15.27$, $p \leq 0.001$). However, the data showed that there are no significant differences among IC and IS employees with regard to turnover intentions (IC = 2.62 vs. IS = 2.75, $F = 0.69$, NS). Generally, these results and corresponding predictor variables are consistent with Table 3, after controlling for all the predictors. Note that no changes occurred when we controlled for the predictor variables. Again, no significant differences were found regarding intention to leave. Given the significant differences in organizational tenure, boundary spanning activities, the two role stressors, overall job satisfaction and organizational commitment among IC and IS employees, separate regression analyses were conducted for IC and IS employees to assess the determinants of turnover intentions. In the next section, we discuss the results associated with testing the path analytical model.

Results for Testing the Path Analytic Model

Tables 4, 5, 6, and 7 present the results of the analyses predicting role stressors, overall job satisfaction, organizational commitment, and turnover intentions, respectively. Note that the absence of an effect among IC employees may be due to the lower statistical power obtained in our sample.

Role Stressors. The model predicted that boundary spanning activities would have direct effects on role ambiguity and role conflict. Table 4 shows that effects of boundary spanning activities on role ambiguity differed significantly. While, for IS personnel, boundary spanning activities had a negative effect on role ambiguity ($\beta = -0.18$, $p \leq 0.05$), its effect on role ambiguity among IC employees was nonsignificant (β

TABLE 2
Relationship of Type of Employee (IC or IS) with the Study Variables

(A) Results of Chi-Square†			
	IC (n = 76)	IS (n = 133)	Chi-Square (χ^2)
Demographic Variables:			
Gender			1.02
Female	40	59	
Male	36	74	
Education			3.50
High school	16	27	
Bachelor's degree	53	81	
Graduate degree	7	25	
(B) Results of ANOVA			
	IC (n = 76)	IS (n = 133)	Univariate F
Demographic Variables:			
Age‡	31.62	30.90	0.25
Organizational Tenure‡	2.40	3.98	18.76***
Boundary Spanning Activities	2.69	1.90	86.87***
Role Stressors:			
Role Ambiguity	3.31	3.03	4.24*
Role Conflict	4.35	3.85	14.41***
Career Success Outcomes:			
Overall Job Satisfaction	1.54	1.91	37.97***
Organizational Commitment	3.94	4.57	15.27***
Intention to Leave	2.65	2.75	0.69

† The values presented in this table are the joint frequency distribution.

‡ We took the log of age and organizational tenure because it is not normally distributed. The values in this table represent the re-transformation of the logarithmic values to number of years.

* $p \leq 0.05$.

*** $p \leq 0.001$.

= 0.09, NS). The demographic variables (e.g., age and education) were found to have strong direct effects on role ambiguity among IC employees. Age was found to have a negative effect on role ambiguity ($\beta = -0.28$, $p \leq 0.05$), but was not significant for IS employees ($\beta = 0.02$, NS). Among IC personnel, education was found to have a positive effect on role ambiguity ($\beta = 0.23$, $p \leq 0.05$) but it showed no significant effect on role ambiguity among IS employees ($\beta = 0.11$, NS). On the other hand, organizational tenure was found to have a negative effect on role ambiguity among both groups, but it was significant only among IS employees ($\beta = -0.19$, $p \leq 0.05$). Only age was found to have a significant effect on role conflict among IS employees ($\beta = -0.19$, $p \leq .05$). All the remaining effects of demographic variables on role conflict among both IC and IS employees were not found to be significant.

Table 4 also shows that the demographic variables and boundary spanning cumulatively explained 18 ($R^2 = 0.18$, $p \leq 0.05$) and 6 ($R^2 = 0.06$, NS) percent of the

TABLE 3
Results of ANCOVA for Study Variables

Variables	Subgroup Means*		Univariate F for Employee Type
	IC	IS	
Boundary Spanning Activities	2.67	1.92	71.62***
Role Stressors:			
Role Ambiguity	3.36	2.98	5.71*
Role Conflict	4.34	3.86	12.76***
Career Success Outcomes:			
Overall Job Satisfaction	1.52	1.90	36.76***
Organizational Commitment	3.99	4.52	10.57***
Intention to Leave	2.56	2.75	1.95

* Means adjusted for covariates. (We controlled for the demographic variables as well as for the corresponding predictors.)

* $p \leq 0.05$.

*** $p \leq 0.001$.

variance in role ambiguity among IC and IS employees, respectively. The effects of demographic variables on role conflict for IC and IS personnel are not significant.

Overall Job Satisfaction. The model posited direct effects of role stressors and boundary spanning activities on overall job satisfaction. As shown in Table 5, role stressors (role ambiguity and role conflict) generally had strong negative effects on overall job satisfaction ($(\beta = -0.40, -0.34, p \leq 0.001$ for role ambiguity among IC and IS employees, respectively), and $(\beta = -0.52, -0.34, p \leq 0.001$ for role conflict among IC and IS employees, respectively)). Boundary spanning activities had a positive effect on overall job satisfaction among IS employees ($\beta = 0.14, p \leq 0.05$). Table 5 also shows that none of the demographic variables had direct effects on overall satisfaction.

The data in Table 5 show that the demographic variables, boundary spanning activities and role stressors, collectively, explained 54 and 37 percent of the variance of overall job satisfaction among IC and IS employees, ($R^2 = 0.54$ and $0.37, p \leq 0.001$, respectively). With the role stressors accounting for 41 and 30 percent ($\Delta R^2 = 0.41$ and $0.30, p \leq 0.001$) percent of the explained variables.

Organizational Commitment. The data regarding the prediction of organizational commitment are shown in Table 6. Overall job satisfaction had a strong positive effect on organizational commitment among both IC and IS employees ($\beta = 0.64$ and $0.48, p \leq 0.001$, respectively). The remaining predictors of organizational commitment among IC employees were found to be nonsignificant. On the other hand, while education ($\beta = -0.19, p \leq 0.05$), organizational tenure ($\beta = -0.15, p \leq 0.05$), and role ambiguity ($\beta = -0.16, p \leq 0.05$) were found to have significant negative effects, boundary spanning activities ($\beta = 0.27, p \leq 0.001$) had a strong positive effect on organizational commitment among IS employees. Table 6 also shows that education had a stronger effect on organizational commitment for IS employees than for IC employees.

Table 6 also shows that role stressors had strong indirect effects through overall job

TABLE 4
Results of Hierarchical Regression Analyses Predicting Role Stressors

(A) Role Ambiguity						
Independent Variables	IC Employees			IS Employees		
	Direct Effect	<i>r</i>	Unexplained*	Direct Effect	<i>r</i>	Unexplained*
Demographic Variables:						
Age	-0.28*	-0.36***	-0.08	0.02 ^a	0.04	0.02
Gender (1 = Female; 2 = Male)	0.06	0.08	0.02	0.06	0.05	-0.01
Education	0.23*	0.20*	-0.03	0.11	0.05	-0.06
Organizational Tenure	-0.18	-0.19*	-0.01	-0.19*	-0.15*	0.04
Boundary Spanning Activities	0.09	0.05	-0.04	-0.18* ^a	-0.14*	0.04
Total <i>R</i> ²	0.18*			0.06		
(B) Role Conflict						
Demographic Variables:						
Age	-0.02	-0.03	-0.01	-0.19*	-0.19*	0.00
Gender (1 = Female; 2 = Male)	-0.01	-0.04	-0.03	0.13	0.13	0.00
Education	0.07	0.06	-0.01	-0.06	-0.05	0.01
Organizational Tenure	-0.15	-0.16	-0.01	0.03	0.10	0.07
Boundary Spanning Activities	-0.09	-0.11	-0.02	0.06	0.07	0.01
Total <i>R</i> ²	0.04			0.06		

* The unexplained effect is the difference between the simple *r* and the sum of the direct, indirect and common cause effects.

* $p \leq 0.05$.

** $p \leq 0.01$.

*** $p \leq 0.001$.

^a Beta weights for IC and IS employees are significantly different ($p \leq 0.05$ or better).

satisfaction among both IC and IS employees. Actually, the effects of role stressors on organizational commitment are mainly indirect via overall job satisfaction. Boundary spanning activities were also found to have a strong indirect effect on organizational commitment among IS employees mainly through overall job satisfaction and role ambiguity.

The model explained 34 and 48 percent ($R^2 = 0.34$ and 0.48 , $p \leq 0.001$) of the variance in organizational commitment among IC and IS employees, respectively. Overall job satisfaction accounted for 19 and 15 percent ($\Delta R^2 = 0.19$ and 0.15 , $p \leq 0.001$) of the explained variance, role stressors 12 and 13 percent ($\Delta R^2 = 0.12$, $p \leq 0.05$ and 0.13 , $p \leq 0.001$), boundary spanning activities 2 and 14 percent ($\Delta R^2 = 0.02$, NS and 0.13 , $p \leq 0.001$), and demographic variables the rest ($\Delta R^2 = 0.01$, NS and 0.07 , $p \leq 0.05$) among IC and IS employees, respectively.

Turnover Intentions. The results of the path model predicting turnover intentions are presented in Table 7. Organizational commitment, as predicted, had a negative effect on turnover intentions among IC and IS employees ($\beta = -0.26$, $p \leq 0.05$, and β

TABLE 5
Results of Hierarchical Regression Analyses Predicting Overall Job Satisfaction

Independent Variables	IC Employees				IS Employees			
	Direct Effect	Indirect Effect	Common Cause	<i>r</i>	Unexplained*	Direct Effect	Indirect Effect	Common Cause
Demographic Variables:								
Age	0.06	0.12	0.00	0.25**	0.07	0.02	-0.06	0.00
Gender (1 = Female; 2 = Male)	-0.03	-0.02	0.00	-0.09	-0.04	-0.02	-0.06	0.00
Education	-0.01	-0.12	0.00	-0.06	0.07	-0.08	-0.02	0.00
Organizational Tenure	0.08	0.14	0.00	0.28**	0.06	0.11	0.05	0.00
Boundary Spanning Activities	0.01	0.01	0.00	0.06	0.04	0.14*	0.04	0.00
Role Stressors:								
Role Ambiguity	-0.40***	0.00	0.01	-0.57***	-0.18	-0.34***	0.00	-0.01
Role Conflict	-0.52***	0.00	0.01	-0.58***	-0.07	-0.34***	0.00	0.00
Total <i>R</i> ²	0.54***					0.37***		

* The unexplained effect is the difference between the simple *r* and the sum of the direct, indirect and common cause effects.

• $p \leq 0.05$.

•• $p \leq 0.01$.

••• $p \leq 0.001$.

TABLE 6
Results of Hierarchical Regression Analyses Predicting Organizational Commitment

Independent Variables	IC Employees				IS Employees			
	Direct Effect	Indirect Effect	Common Cause	<i>r</i>	Unexplained*	Direct Effect	Indirect Effect	Common Cause
Demographic Variables:								
Age	-0.16	0.12	0.00	0.03	0.07	0.08 ^a	-0.02	0.00
Gender (1 = Female; 2 = Male)	0.06	-0.03	0.00	0.05	0.02	-0.08	-0.04	0.00
Education	0.06	-0.15	0.00	-0.04	0.05	-0.19 ^{aa}	-0.07	0.00
Organizational Tenure	-0.12	0.14	0.00	0.01	-0.01	-0.15 ^a	0.09	0.00
Boundary Spanning Activities	0.16	0.00	0.00	0.14	-0.02	0.27 ^{***}	0.12	0.00
Role Stressors:								
Role Ambiguity	-0.03	-0.26	0.10	-0.27 ^{***}	-0.08	-0.16 ^a	-0.16	-0.05
Role Conflict	0.11	-0.33	0.01	-0.26 ^{***}	-0.05	0.07	-0.16	0.03
Overall Job Satisfaction	0.64 ^{***}	0.00	-0.10	0.54 ^{***}	0.00	0.48 ^{***}	0.00	0.10
Total <i>R</i> ²	0.34 ^{***}					0.48 ^{***}		
								Unexplained*
								-0.05
								0.03
								0.08
								-0.06
								-0.05
								-0.06
								-0.12
								0.00

* The unexplained effect is the difference between the simple *r* and the sum of the direct, indirect and common cause effects.

^a $p \leq 0.05$.

^{aa} $p \leq 0.01$.

^{***} $p \leq 0.001$.

^a Beta weights for IC and IS employees are significantly different ($p \leq 0.05$ or better).

TABLE 7
Results of Hierarchical Regression Analyses Predicting Intention to Leave

Independent Variables	IC Employees				IS Employees					
	Direct Effect	Indirect Effect	Common Cause	<i>r</i>	Unexplained*	Direct Effect	Indirect Effect	Common Cause	<i>r</i>	Unexplained*
Demographic Variables:										
Age	-0.29*	-0.09	0.00	-0.22*	0.16	-0.11	-0.01	0.00	-0.08	0.04
Gender (1 = Female; 2 = Male)	-0.17	0.03	0.00	-0.09	0.05	0.01	0.09	0.00	0.07	-0.03
Education	-0.07	0.09	0.00	-0.03	-0.05	0.05	0.17	0.00	0.12	-0.10
Organizational Tenure	-0.32**	0.10	0.00	-0.28**	-0.06	-0.02 ^a	0.01	0.00	-0.06	-0.05
Boundary Spanning Activities	0.13	-0.02	0.00	0.12	0.01	-0.06	-0.26	0.00	-0.29***	0.03
Role Stressors:										
Role Ambiguity	0.14	0.22	0.12	0.36***	-0.12	0.01	0.23	0.02	0.34***	0.08
Role Conflict	-0.10	0.25	0.04	0.18*	-0.01	0.05	0.09	-0.03	0.18*	0.07
Overall Job Satisfaction	-0.37*	-0.17	-0.11	-0.48***	0.17	-0.08 ^a	-0.30	-0.04	-0.58***	-0.16
Organizational Commitment	-0.26*	0.00	-0.23	-0.49***	0.00	-0.63*** ^a	0.00	-0.09	-0.72***	0.00
Total <i>R</i> ²	0.44***					0.54***				

* The unexplained effect is the difference between the simple *r* and the sum of the direct, indirect, and common cause effects.
• $p \leq 0.05$.
** $p \leq 0.01$.
*** $p \leq 0.001$.
^a Beta weights for IC and IS employees are significantly different ($p \leq 0.05$ or better).

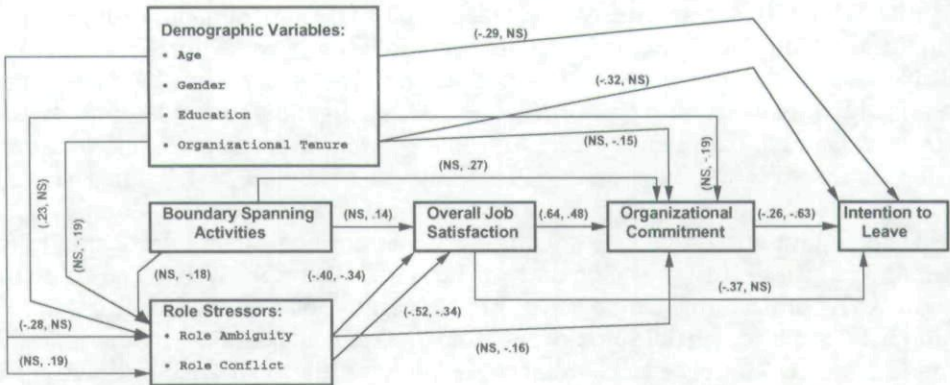
= -0.63, $p \leq 0.001$, respectively). Organizational commitment had a stronger effect on turnover intentions among IS employees than IC employees. On the other hand, Table 7 shows that overall job satisfaction had a direct effect on turnover intentions among IC employees only ($\beta = -0.37$, $p \leq 0.05$), and its effect among IC employees is stronger than among IS employees. Actually, overall job satisfaction had no direct effect on turnover intentions among IS employees ($\beta = -0.08$, NS), but had a strong indirect effect through organizational commitment.

Table 7 shows that age and organizational tenure had direct effects among IC employees ($\beta = -0.29$, $p \leq 0.05$ and -0.32 , $p \leq 0.01$, respectively). The data also show that organizational tenure had a stronger negative effect on turnover intentions among IC employees than IS employees. Role stressors and boundary spanning activities had mainly indirect effects on turnover intentions through organizational commitment and overall job satisfaction among both IC and IS employees.

The data show that the study variables cumulatively explained 44 and 54 percent ($R^2 = 0.44$ and 0.54 , $p \leq 0.001$) of the explained variance of turnover intentions, with organizational commitment accounting for 5 and 21 percent of the explained variance ($\Delta R^2 = 0.05$, $p \leq 0.05$ and 0.21 , $p \leq 0.001$), overall job satisfaction for 13 and 10 percent ($\Delta R^2 = 0.13$, $p \leq 0.001$ and 0.10 , $p \leq 0.001$), role stressors for 14 and 9 percent ($\Delta R^2 = 0.14$, $p \leq 0.01$ and 0.09 , $p \leq 0.001$), boundary spanning activities for 1 and 9 percent ($\Delta R^2 = 0.01$, NS and 0.09 , $p \leq 0.001$), and demographic variables the rest 11 and 5 percent ($\Delta R^2 = 0.11$ and 0.05 , NS), among IC and IS employees, respectively.

The large common cause effect (organizational commitment with turnover intentions among IC employees, -0.23) indicates that the simple correlation between these constructs are largely due to other, causally prior variables. Overall job satisfaction and role stressors are the major common cause of the large correlation between organizational commitment and turnover intentions ($r = -0.49$, $p \leq 0.001$).

Finally, applying the criterion that the absolute differences between the observed and recomputed correlations do not exceed 0.10 (Namboodiri, Carter and Blalock 1975), the data showed that the reduced model duplicated all 20 correlations with role stressors (role ambiguity and role conflict among IC and IS employees). For overall job satisfaction, 12 of the 14 reconstructed correlations were consistent with the original coefficients. For organizational commitment, 15 out of 16 reconstructed correlations fell within the limits specified. Lastly, for turnover intentions, Table 7 shows that 14 out of 18 reconstructed correlations were consistent with the original coefficients. In summary, the effects hypothesized but not explained by the model are generally small, as shown in Tables 4, 5, 6, and 7. Among the 68 effects implied to be zero, 7 are strong enough to note. The strong unexplained effects among IC employees are: overall job satisfaction with turnover intentions, 0.17; role ambiguity with turnover intentions, -0.12 ; age with turnover intentions, 0.16; and role ambiguity with overall job satisfaction, -0.18 . The strong unexplained effects among IS employees are: overall job satisfaction with turnover intentions, -0.16 ; role conflict with organizational commitment, -0.12 ; and role ambiguity with overall job satisfaction, -0.14 . There are strong correlations between some of the variables among IC and IS employees, respectively: overall job satisfaction and turnover intentions, -0.48 and -0.58 ; organizational commitment and turnover intention, -0.49 and -0.72 ; role ambiguity and turnover intentions, 0.36 and 0.34; role ambiguity and overall job satisfaction, -0.57 , and -0.49 ; role conflict and overall job satisfaction, -0.58 and



Note. The numbers in parentheses are the path coefficients of the predictor variables among IC and IS employees, respectively.

FIGURE 2. Path Diagram of the Effects of the Exogenous Variables on the Endogenous Variables.

-0.44; overall job satisfaction and organizational commitment, 0.54 and 0.58. These strong correlations call for further investigation of these important variables.

In summary, Tables 2 and 3 present the chi-square tests, the analysis of variance and covariance for the study variables. Tables 4, 5, 6, and 7 present the results of the hierarchical multiple regression analyses for the fully recursive model, and indicate the direct, indirect and common cause effects, the simpler of the causal variables on the dependent variables, and the unexplained effect. The data provide partial support for the overall model proposed in Figure 1. Figure 2 summarizes the results for the proposed model.

Implications for IS/IC Management

The results corroborate much of the findings from previous studies discussed earlier. Employee intention to leave the organization is found to be strongly influenced by several factors: employee commitment to the organization, overall job satisfaction and some employee characteristics such as age and length of time with the organization. In turn, employee commitment to the organization is directly affected by overall job satisfaction, role ambiguity, the level of employee boundary spanning activities, employee tenure with the organization and level of education. Overall job satisfaction is directly affected by the level of employee boundary spanning activity, role ambiguity and role conflict.

This section addresses the study's findings in terms of their implications and related practical recommendations for IC and IS managers. The operational significance of the statistical results presented earlier should be considered promising because statistically significant results based on sample sizes of 50 or 60 "are probably also of practical significance" (Dunnette 1966, p. 156). Further, the statistical power of both samples in this study exceeds 0.50. In many cases, the recommendations already have some precedent in practice. We caution the reader that some of the recommendations may have unintended side effects. Whenever possible we pointed these out. The study found little evidence that boundary spanning activities among IC employees would generate extensive role stress and lower levels of overall job

satisfaction and organizational commitment. This may be due to the fact that IC employees are psychologically better prepared to handle more extensive boundary spanning activities than IS employees. That is considered a vital part of their job (Christoff 1990; Perry 1987). Accordingly, IC employees very likely have been hired for their interpersonal and communications skills so necessary to cross departmental and organizational boundaries without generating role stress. On the other hand, IS employees have a more technical background with relatively less emphasis on people skills. This coincides with the motivation for the establishment of the well-known "user liaison" role to bridge the traditional communication gap between IS and end-user departments.

Among IC personnel, personal maturity (age) shows a significant inverse effect on role ambiguity. This may be a clue for IC managers to watch their younger subordinates more carefully, given that IC personnel report significantly higher levels of role ambiguity than their IS counterparts. Apparently, older IC employees have learned to manage the high levels of role ambiguity. IC managers must also ensure that younger members of the IC organization are assigned jobs with reduced levels of role ambiguity. Possible measures to reduce role ambiguity are discussed later. Alternatively, IC managers may view high role ambiguity as an integral part of a "sink or swim" approach to training and weeding out of young IC personnel. However, such an approach is likely to be counterproductive due to high employee anxiety, frustration, and political fallout from the weeding-out process.

Inconsistent with Baroudi's finding (1985), the degree of role ambiguity is affected by IS employee's level of boundary spanning activity. While IS personnel have reported significantly lower levels of department boundary spanning, such activities have an inverse effect on the role ambiguity associated with their job. A possible explanation is based on the fact that IS tasks (systems development) are relatively well defined and understood, and benefit from a wide collection of formal methodologies, tools and techniques (some of which are computerized in CASE). In such cases, crossing departmental boundaries is facilitated by the relatively high degree of task structure and definition. If this interpretation is correct, this finding suggests to IC managers the need to define support roles clearly and for adopting or developing equivalent methodologies, tools and techniques to support end-user computing and reduce role ambiguity. Preliminary evidence that methodologies can enhance the communication process between various participants in systems development has been found by Yoon and Guimaraes (1992).

Employee educational level does not have the same helpful influence on role ambiguity as their personal maturity. To the contrary, higher education levels are directly related to higher levels of role ambiguity for IC personnel. Some other factors must be considered in this case. On the average IC personnel reported significantly shorter time working for the organization; therefore, one may assume many of them are not coming to the IC via the company's IS department. Carr (1988, p. 96) also found that IC employees have shorter organizational tenure than their IS counterparts. There is evidence that end-user departments in many companies are informally implementing their own ICs given the inability of central ICs to keep up with the workload (Crwth 1991, p. 13). Possibly the "average" IC professional is relatively fresh out of school and is expected to possess considerable skills on personal computing, business administration, as well as strong interpersonal communication abilities. They are not expected to know methodologies, tools and techniques which provide so much struc-

ture to systems development tasks, and are likely to provide an advantage for IS professionals in reducing role ambiguity. The previous recommendations for reducing role ambiguity are, of course, also applicable here.

Regardless of its source, IC managers can reduce role ambiguity by taking steps to ensure (1) more clear definition of tasks and task priorities related to the job, (2) better employee understanding of the task objective and the reasons for performing the task, (3) better task design and task definition so the beginning and completion of each task are readily determined, (4) more clearly defined sequence in which subtasks should be performed, and (5) a better understanding of how a particular task should be performed. It behooves IC managers to consider adopting some of the methods, tools and procedures used in other departments to assist IC personnel in dealing with the higher levels of role ambiguity associated with end-user computing support.

IC personnel on the average report substantially higher levels of role conflict than their IS counterparts. Employee age has an inverse effect on role conflict, and employee organizational tenure has an inverse effect on role ambiguity among IS employees, but the data show no statistically significant difference between IC and IS personnel. Apparently employee personal maturity and familiarity with the organization's ways are quite helpful in handling causes of role stress (role conflict and ambiguity): knowing how/when to act under different conditions, borrowing appropriate resources to complete assignments, effectively working with different groups of people, resolving incompatible requests, etc. This finding suggests that organizations should carefully consider ways to help younger and/or new employees manage role stress which has a strong negative impact on job satisfaction for IC and IS employees alike. Some useful measures may include orientation sessions for new employees on organization structure, organization formal and informal resource allocation procedures, and what to do in certain types of commonly encountered stress situations. Some large, complex organizations (i.e., General Mills) have successfully implemented a "buddy system" where a more senior employee adopts a new (and usually younger) employee and provides direction and assistance with a wide variety of questions.

For IC groups operating within IS departments, managers should hold coordination meetings to facilitate the sharing of development ideas between the IS and IC personnel. These meetings will help managers keep track of the end-user applications being developed jointly. These will also help the IC employees to be up to date and resolve conflicting demands from end-users and IS department. Finally, managers should clearly define the responsibilities of IC, IS and end-user personnel participating in a system development effort to have clearly defined tasks and also enhance cooperation between IS and IC employees providing end-user support. These measures should help reduce stress caused by role ambiguity and conflict.

IC personnel, on the average, report significantly lower levels of job satisfaction than their IS counterparts. However, the levels of role ambiguity and role conflict have strong direct effects on the overall job satisfaction of both IC and IS personnel. The higher levels of role conflict and role ambiguity reported by IC personnel may explain the stronger relationships to job satisfaction in their case; however, the differences are not statistically significant. Nevertheless, there is a strong suggestion that as the levels of role conflict and ambiguity grow, so do their effects on employee overall job satisfaction. Therefore, managers in both areas should give serious consideration

to implementing some of the measures to reduce role ambiguity and conflict enumerated earlier.

The relationships between role stressors and overall job satisfaction were completely supported among both IC and IS employees. Role ambiguity and role conflict have shown a strong inverse relation to overall job satisfaction. This is consistent with the findings of Bedeian and Armenakis (1981), Brief and Aldag (1980) and others. It is also consistent with Baroudi's findings (Baroudi 1985), and suggests that role stressors deserve a great deal of attention from IC and IS managers. The findings support the notion that the impact of role stressors on turnover intentions is mediated by overall job satisfaction and organizational commitment. IC and IS managers can substantially increase overall job satisfaction levels, increase employee commitment and reduce employee's intention to leave the organization by reducing the level of role ambiguity and role conflict. Managers must ensure that the levels of role conflict and ambiguity are not raised unnecessarily as a result of poor management practices and policies within the department. Furthermore, since role conflict and role ambiguity may be conditions germane to the IC environment, IC managers should pay special attention to recruiting employees who have relatively high tolerance for both role conflict and role ambiguity.

The reported level of boundary spanning activities for IC personnel is significantly higher but seems to have no direct effect on employee job satisfaction. This may be due to insufficient statistical power in the IC personnel sample. On the other hand, contrary to Baroudi's findings, the lower level of boundary spanning activities for IS personnel apparently improves job satisfaction. A possible explanation is that with the great increase in the importance of end-user computing activities for most companies, IS personnel view interaction with people in other departments as a desirable thing, perhaps an important part of career development. This finding complements the point discussed earlier that, given the wealth of methodologies, tools and procedures for guiding IS personnel through systems development, boundary spanning activities have an inverse effect on role ambiguity, thus indirectly improving overall job satisfaction.

The results of this study support the notion that overall job satisfaction is the primary motivator for IC and IS employees to be organizationally committed. Overall job satisfaction has a direct influence on one's commitment to the organization and the intention to leave it, among IC employees. However, it had no direct effect on turnover intentions among IS employees, which is consistent with Baroudi's (1985) findings. As the major determinant of employee intention to leave the organization, commitment to the organization is worthy of careful attention. Several factors directly affect IS employee commitment: the level of role ambiguity, overall job satisfaction, the level of boundary spanning activities, employee education and organizational tenure. Despite the higher levels of role ambiguity associated with IC personnel, no significant effect on their commitment to the organization is apparent. However, for IS personnel, higher levels of role ambiguity lead to lower commitment. Apparently, IS professionals are less tolerant of what they perceive as unreasonable levels of role ambiguity. This is further evidence that managers, particularly managers of IS personnel, should readily implement some of the suggested role ambiguity reduction measures discussed earlier. Among IS personnel, education and organization tenure have an inverse effect on employee commitment to the organization. One

possible explanation is that the more highly educated and experienced employees are, the more conscious they become of their value and marketability. Another, perhaps complimentary, explanation is that after a few years with one organization, more educated people long for a new environment and/or a new job to further enrich their experience and knowledge. The strongest impact on employee commitment to the organization (which is significantly lower for IC personnel) comes from overall job satisfaction. Given its direct and indirect effects on employee turnover intentions, managers should carefully consider doing whatever is possible to measure, monitor and control employee job satisfaction and its determinants.

The earlier findings (Baroudi 1985; Bartol 1983; Dougherty et al. 1985; Michaels and Spector 1982) that organizational commitment has a direct effect on employees' intention to leave have been corroborated. Employee intentions to leave the organization are affected by organizational commitment, overall job satisfaction, employee age and organization tenure. Except for organizational commitment, these determinants of employee intentions to leave are significant only for IC personnel. Older IC employees, on the average, report a lower level of intentions to leave the organization. This should come as good news to IC managers since IC employee age was found to have a significant inverse effect on role ambiguity, a major factor to overall job satisfaction.

For IS employees, job satisfaction does not have nearly as much impact on turnover intentions, as for their IC counterparts. Consistent with Baroudi's findings (1985), the level of IS personnel commitment to the organization is the key determinant of intention to leave. Therefore, IS managers should implement measures more energetically that increase employee organizational commitment: try harder to understand employee values, needs and motivation; build company and department esprit de corps; make them feel proud of working for the organization.

On the other hand, IC managers should focus on job satisfaction, particularly for younger employees and those with less tenure in the organization. Besides implementing the measures discussed earlier regarding the determinants of job satisfaction, managers should try to specialize employees in doing tasks that they are better at or enjoy doing. It is particularly important that IC managers try to ensure employee satisfaction with their work, their supervisors, the people they work with. They should also strive for competitive job remuneration, fair promotion policies, and attractive career paths. IC managers might improve employee commitment to the organization and reduce their intention to leave by addressing the image problem associated with IC roles. Rainer et al. (1989) have found that IC professionals perceive their jobs as lacking in career pathing and promotion. Very likely therein lies a major reason for their lower commitment to the organization. Perhaps IC career pathing should include certain positions outside the IC, such as user liaison, systems analyst, and staff or line positions within user departments, which with individual IC employees are most familiar. Substantial retraining will be necessary to improve career pathing effectively for IC personnel. Further research is needed to examine in more detail these recommendations regarding job satisfaction among IC employees. However, results from this study strongly suggest that organizations where IC managers are careless about employee job satisfaction represent an apparently easy target for raiding organizations and "head hunters." In this case, the raiding organizations can be quite successful if, in turn, they would manage to maintain adequate employee job satisfaction.

Study Limitations and Directions for Future Research

While the present study has corroborated the findings of many of the studies discussed earlier, provided moderate support for the hypotheses and mediational processes proposed in the model, and provided some important insights for managing IC and IS professionals, it also raises several issues which represent substantial opportunities for research.

(1) We need future research encompassing a wider sample of IC and IS employees than the present convenience sample, in order to ensure the applicability of the findings to the general population of IS and IC employees. Specifically, we need a larger sample of IC employees for greater statistical power.

(2) Future research should focus on the impact of increased boundary spanning activities among IS and IC people. The trend within some companies to move end-user computing support from centralized IC to IS and end-user departments will change the role of end-users, IC and IS employees. Future research should examine the impact of these and other IC organization changes on the level of boundary spanning, and the managerial implications.

(3) Longitudinal studies involving multiple data points should be conducted to capture the dynamic processes involved in turnover behavior more completely, and better assess the impact of job satisfaction on turnover itself, not only on employee's intentions to leave the organization.

(4) Future models should include a wide variety of variables such as job characteristics (performance level and organizational rewards), organizational characteristics (number of employees, proximity of services, budget percentages allocated to IC and IS, whether the IC is located within the IS department, within the user department, or is a separate department), and the level of support provided to the end-user community.

(5) The career paths of both IC and IS employees and the impact of cross-functional movement of personnel on turnover needs study.

(6) There is need for research to examine differences in career orientations of IC and IS employees, the relationships between their career orientations and type of positions both these employees hold in their career paths, and how these differences in career orientations affect employee turnover.

(7) A recent article (Li and Shani 1991) confirms that managerial action can reduce role ambiguity and role conflict. It is important that this confirmation process be continued and broadened to include other variables. The ultimate test for any model is its ability to produce desired changes in an applied setting. Therefore, future research should examine whether all the specific managerial actions recommended here indeed produce the expected effects. Research programs that combine theoretically oriented studies on the turnover process, using second generational multivariate analysis with field-based, quasi-experimental interventions should provide major insights of both a theoretical and practical nature.

(8) Although common method variance due to self-reporting bias is not found to be a problem in this study, researchers must strive to develop more direct and objective measures for the variables being studied, to reduce the likelihood of obtaining spurious results in the future. The following evidence suggests that the results presented here are relatively bias free: First, Table 1 shows no tendency for respondents to agree/disagree with items, regardless of content. Second, correlations between

objective measures (demographic variables) and others are consistent with expectations based on previous studies and experience. Third, the risk of common method variance is much greater for single-item measures or poorly designed scales (Spector 1987). This study is based on previously tested multi-item scales with adequate reliability.

(9) Last, the increasingly global view that information resources managers are being forced to assume, and the increasingly broader definition of end-user to include customers, suppliers, and even competitors, intensify the importance of boundary spanning activities as a variable. A worthwhile research effort is to pre-empt the need for IS managers to understand and manage the impact of boundary spanning in this global and deeper context where a multicultural end-user community is likely to provide new challenges to IS researchers, as well as practitioners.

Despite the above limitations, this article addresses an exceedingly important issue: the human element behind information technology, specifically, the differences between the IC and IS employee groups. Based on an adaptation of Baroudi's model (1985), the relationships among the determinants of turnover intentions for IC and IS people were examined in a comparative fashion. The results provide important insights into the role of demographic variables, boundary spanning activities, role stressors, overall job satisfaction and organizational commitment in influencing, both directly and indirectly, turnover intentions among IC and IS employees.*

Acknowledgements. The authors are grateful to the anonymous Associate Editor and three reviewers for their incisive, thorough and valuable comments, and to George Huber, Jack Baroudi and Detmar Straub whose advice also helped improve earlier versions of this paper. We thank Teresa Flood and Glenna Groce for their assistance, and the IC/IS professionals for contributing their time and experience to this study.

* Omar El Sawy, Associate Editor. This paper was received on August 30, 1990, and has been with the authors 6½ months for 3 revisions.

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